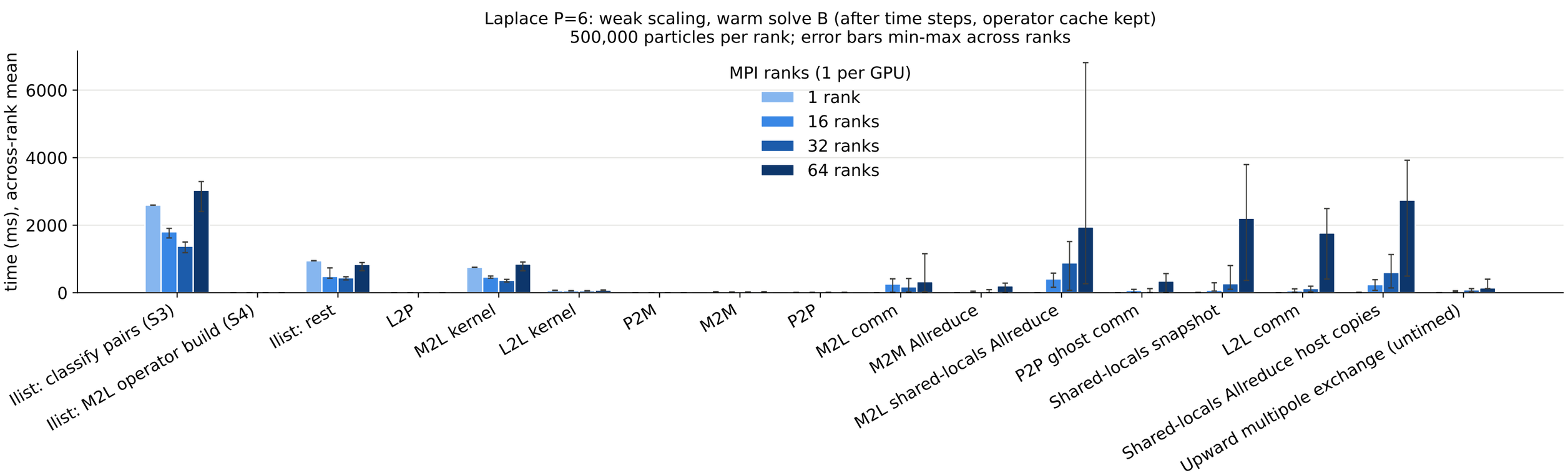
Laplace P=6: weak scaling, warm solve B (after time steps, operator cache kept)
500,000 particles per rank; error bars min-max across ranks
time (ms), across-rank mean
0
2000
4000
6000
MPI ranks (1 per GPU)
1 rank
16 ranks
32 ranks
64 ranks
Ilist: classify pairs (S3)
Ilist: M2L operator build (S4)
Ilist: rest
L2P
M2L kernel
L2L kernel
P2M
M2M
P2P
M2L comm
M2M Allreduce
M2L shared-locals Allreduce
P2P ghost comm
Shared-locals snapshot
L2L comm
Shared-locals Allreduce host copies
Upward multipole exchange (untimed)